

Contract-based Agentic Intent Framework for Network Slicing in O-RAN

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Abstract—Intent-based networking aims to simplify network operation by translating operator intents into a collection of policies, configurations, and control actions. However, this translation process relies on heuristics and loose coupling. It often results in unpredictable behavior and ambiguous safety standards. This paper presents a Contract-based Agentic Intent Framework (CAIF) for the radio access network (RAN). The proposed framework employs a closed-loop agentic pipeline that systematically audits user objectives against formal RAN constraints prior to actuation. The proposed CAIF decouples probabilistic intent extraction from strictly governed policy execution to enable the enforcement of deterministic safety guarantees. We use network slicing as a representative use case to demonstrate the design flow and validate the effectiveness of the proposed approach on an O-RAN testbed. Experimental results show that the closed-loop agentic pipeline of the proposed CAIF can effectively eliminate harmful intent executions observed in direct-actuation baseline approaches.

Index Terms—intent-based network, agentic AI, LLM, Open RAN

I. INTRODUCTION

The transition to 6G networks introduces an era of network flexibility, where the management of heterogeneous resources like Radio Resource Management (RRM) requires a fundamental shift toward full autonomy. Intent-Based Networking (IBN) has emerged as the essential paradigm, allowing operators to define high-level operational goals without managing low-level parameters. This autonomy is particularly critical for Network Slicing [1], a complex mechanism requiring the dynamic, cross-domain partitioning of resources to meet strict Service Level Agreements (SLAs). However, realizing the full potential of IBN for slicing remains challenging due to the rigid nature of traditional translation models. In this context, Large Language Models (LLMs) present a transformative opportunity, offering a cognitive layer capable of interpreting and translating high-level human intent and decomposing it into actionable logic [2] [3].

Translating a high-level goal into a specific outcome requires precise, granular control over network resources. It requires a programmable and modular infrastructure where every component can be individually managed via software. Programmable RAN architectures [4] serve as the necessary execution substrate by decoupling control logic from physical radio hardware through open and standardized interfaces. This separation transforms the network from a monolithic black box into a modular platform where control functions

are exposed as software applications, specifically Non-Real-Time (rApps) and Near-Real-Time (xApps). By utilizing this disaggregated framework, high-level slicing requirements can be programmatically mapped to specific control loops, providing the granular, evidence-based actuation layer necessary to partition radio resources dynamically in a real-world environment.

A prior study implementing an intent-driven mechanism for O-RAN automation demonstrates strong potential in orchestrating multiple xApps to optimize key performance indicators (KPIs) such as throughput and energy efficiency [5]. However, it still rely on manually specified operator intents. The system requires operators to input explicit KPI targets. This approach lacks an intent-interpretation layer capable of translating human-level semantic instructions into machine-readable optimization goals. To address these constraints, we propose the integration of LLMs as a flexible intent-interpretation layer. However, we anticipate that this may introduce architectural risks that require careful mitigation.

Deploying LLM-based agents introduces inherent reliability risks due to their probabilistic nature, where outputs may drift even under static inputs [6]. While methods like JSON schemas enforce formatting [7], they fail to validate decision logic, leaving the network unable to distinguish between valid optimizations and dangerous fabrications. Without strict limits, these agents can react too aggressively to a request, causing a chain reaction of errors that crashes the wider network [8]. Our prior work confirmed this vulnerability, showing that agents often yield inconsistent interpretations despite accurate parsing [9]. To evaluate this problem, we adopt this earlier architecture as our Direct-Actuation Baseline, using it to demonstrate that without a formal Intent Contract as a deterministic guardrail, autonomous decisions remain unsafe to execute.

To address these reliability issues, we present a Contract-based Agentic Intent Framework specifically built for slicing in O-RAN networks. Our contributions are as follows:

- **Contract-based Agentic Intent Framework Architecture:** We design a translation pipeline where the LLM output is strictly governed by a formal Intent Contract which enforces deterministic validation before execution.
- **Slicing Specific App Orchestration:** We demonstrate how high-level intent contracts are systematically decomposed and enforced across a hierarchical rApp and xApp

workflow in a programmable RAN, enabling dynamic and SLA aware network slicing.

- **Implementation in Open RAN Architecture:** We validate the practical feasibility of the proposed framework on an Open RAN testbed, showing how natural language intents are translated into standardized rApp or xApp control policies under contract enforcement.

The manuscript is organized as follows. Section II discusses the proposed architecture. Section III discusses the results of the experiment. Finally, Section IV addresses the lesson learned from our experiment and potential future works.

II. CONTRACT-BASED AGENTIC INTENT FRAMEWORK

This paper proposes a Contract-based Agentic Intent Framework (CAIF) designed to operate within the disaggregated O-RAN architecture. Our method uses Generative AI to understand complex requests. However, to prevent errors, we require the AI to assist in generating a *digital contract* that verifies the safety of its decision before the network executes it. The architecture establishes a modular pipeline where high-level intent is first processed into a formal contract before being executed via standardized O-RAN interfaces.

A. Contract-based Intent Model

Our framework integrates the TMF-921 Intent Management API [10] to create a formal *Intent Contract*. To ensure deterministic execution, we enforce a strict schema where abstract goals are mapped to specific JSON-LD paths. As shown in Table I, the contract mandates specific KPIs that strictly define the agent's operating boundary prior to actuation.

The contract consists of mandatory KPIs that the agentic framework must populate and validate:

- **Scope and Expectations:** The `icm:target` and `icm:hasExpectation` fields explicitly define the scope of the operation. In the example shown, the intent is strictly limited to Throughput Enhancement on `Cell_1_Slice_1`, preventing the agent from inadvertently modifying adjacent resources or applying incorrect optimization logic.
- **Deterministic Constraints:** Unlike open-ended LLM outputs, the contract enforces quantifiable limits. The Target Value (e.g., 5%) sets a rigid boundary for the optimization, ensuring that the `ran:targetThroughputIncrease` does not exceed safe operational thresholds.
- **Policy Mechanism:** The `idan:Delivery_slaPolicy` field specifies the exact control loop to be utilized—in this case, the Two-level RRM Policy Ratio. This decoupling ensures that while the LLM determines what needs to happen, the underlying standardized rApps determine how it is executed, preventing the generation of hallucinated or non-standard command sequences.
- **Governance and Relationships:** To ensure lifecycle integrity, the contract links to a specific `IntentSpecification` (`slaSliceSpec`) and establishes an `IntentRelationship` with the policy-baseline. This hierarchy allows the Intent

Management system to detect conflicts with existing policies before they are activated.

- **Flattened Characteristics:** For rapid indexing and retrieval by rApps/xApps, complex graph attributes are flattened into the characteristic array (e.g., `eligibleClusters`), allowing low-latency components to access vital parameters without parsing the full semantic graph.

B. Agentic AI for Dynamic Translation

CAIF adopts a dual-agent architecture to support dynamic and reliable intent translation. The overall system consists of four core components, which are detailed as follows:

- **Profiling Agent:** Analyzes multi-turn natural language inputs and extracts structured intent information according to a predefined intent model. It determines the intent category and key parameters while ensuring schema compliance.
- **Evaluator Agent:** Validates the extracted intent by checking completeness, consistency, and constraint compliance. If violations or missing fields are detected, it triggers refinement to improve translation reliability.
- **Contract Generator:** Transforms the validated intent into a standardized, machine-readable contract. The generated contract is suitable for execution, validation, and auditing.
- **Intent Management:** Registers and tracks contracts through their entire lifecycle. It also maintains the *IntentSpecifications* that define what the system is allowed to do.

C. Prompt Techniques

We design three types of prompts, namely the Intent Profiling Prompt, Intent Evaluation Prompt, and Intent Refinement Prompt, to achieve reliable and controllable intent extraction using LLMs. The three prompts share a similar structure as shown in Fig. 1. The structure constrains the LLM to the telecommunications domain and applies iterative filling together with strict schema enforcement to translate multi-turn natural language inputs into machine-readable intents. To handle the unpredictability of LLM outputs, the Intent Evaluation Prompt acts as a strict content validator. This mechanism cross-checks the extracted intent against the full conversation history to ensure the accuracy of intent. Following this, the Intent Refinement Prompt triggers a correction loop. Instead of rewriting the entire request, this step fixes only the specific errors identified by the evaluation and preserves the parts of the intent that were already correct.

D. Intent Translation Pipeline

Utilizing the prompt strategies from Sect. II-C, we construct an agentic pipeline to translate semantic intents into contracts. This architecture ensures semantic accuracy through agentic reasoning and iterative self-correction. The system relies on two specialized agents: the Profiling Agent, which handles intent parsing and structuring, and the Evaluator Agent, which performs error detection and semantic review.

TABLE I
MANDATORY KPIS AND CONSTRAINTS DEFINED IN INTENT CONTRACT

KPI / Parameter	Value / Constraint	Source Context
Target Resource	Cell_1_Slice_1	icm:target
Expectations	Throughput Enhancement	icm:hasExpectation
Target Value	5%	ran:targetThroughputIncrease
Policy Mechanism	Two-level RRM Policy Ratio	idan:Delivery_slaPolicy
IntentSpecification	slaSliceSpec	id
IntentRelationship	policy-baseline	id
characteristic	[eligibleClusters, affectedCells, string, Cell_1_Slice_1]	[id, name, valueType, value]

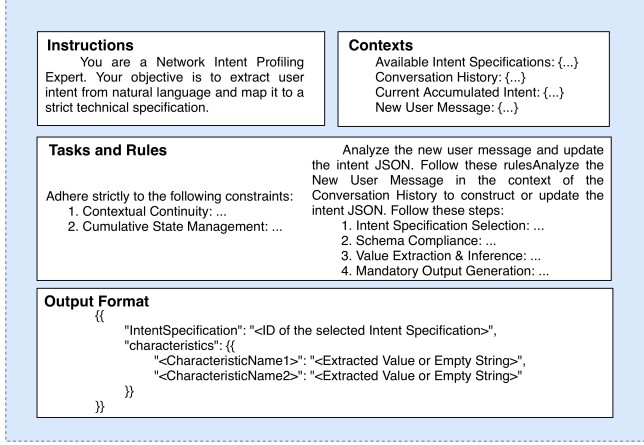


Fig. 1. Intent Profiling Prompt

The workflow starts with the Profiling Agent, which translates the operator's input into a structured format derived from the intent specifications. The Evaluator Agent then inspects this output to detect inconsistencies, hallucinations, or unsupported requests. If anomalies are identified, the system triggers a closed-loop refinement process, guiding the Profiling Agent to rectify the specific discrepancies based on feedback. Once the output is fully validated, the Contract Generator produces the final Intent Contract. This systematic verification ensures that strictly deterministic instructions are enforced prior to network execution.

E. CAIF Implementation on the O-RAN Platform

We implement the CAIF for network slicing within an O-RAN environment, as illustrated in Fig. 2. To ensure a realistic testing environment, our system utilizes an experimental platform similar to the one described in [11]. This setup allows us to demonstrate the practical execution of Intent Contracts through specific rApp and xApp service calls.

1) *SLA Slice and A1 Policy Handler*: Once the SLA Slice rApp receives the formal intent, it converts the user request from a percentage into an absolute throughput threshold. For example, if a user asks for a percentage change, the system calculates exactly how many megabits per second that represents. This calculation uses both live and past network data collected from the O-RAN O1 interface within the SMO.

After setting this target, the SLA Slice rApp performs a feasibility check to see if the network can actually reach that

goal. It does this by looking at a table of past performance that tracks Physical Resource Block (PRB) and Channel Quality Indicator (CQI) of a cell. By comparing the new request to this history, the system decides if the target is possible under the current radio conditions. Only requests that pass this check move forward to the next step in the control process.

After a successful check, the SLA Slice rApp creates the necessary A1 policy data and sends a unique policy name back to the CAIF. Once the A1 Policy Handler rApp receives the notification, it opens and reads the full Intent Contract to find the matching policy data. It then sends this information to the A1 Mediator on the Near-RT RIC using the A1 Policy Management Service within the SMO.

2) *Closed-Loop RAN Slice Control with E2 KPM and RC*: To execute the policies generated by the CAIF, we implement a closed-loop control system within the Near-RT RIC. This workflow relies on the interaction among three specialized xApps to maintain the required slice performance. First, the KPIMON-GO xApp continuously monitors the network by collecting key performance measurements and storing them in a database.

The SLA Slice xApp then retrieves this data to evaluate slice performance against the defined contract. If the current throughput does not meet the goal, this app dynamically calculates new values for the maximum and minimum resource ratios as shown in Fig. 3. These decisions are passed to the RC xApp, which functions as the execution unit. The RC xApp converts these high-level control decisions into encoded messages that are sent to the RAN.

III. EXPERIMENTAL RESULTS

A. Experimental Setup and Methodology

The evaluation of the proposed contract-based framework is conducted using the O-RAN experimental platform and testbed infrastructure established in our previous work [11]. This platform provides the necessary cell configuration and measurement functions to simulate a disaggregated RAN environment. For RAN itself, we utilize the commercial AI RSG RAN simulator from VIAVI. For this study, we focus on the reliability of the intent to policy translation process. We generate a curated and custom dataset of 500 natural language intent instances, with variations across different shot cases ranging from 1 to 5, which is publicly available at [12]. This variation covers a wide range of possibilities for user intent. These instances feature diverse semantic structures

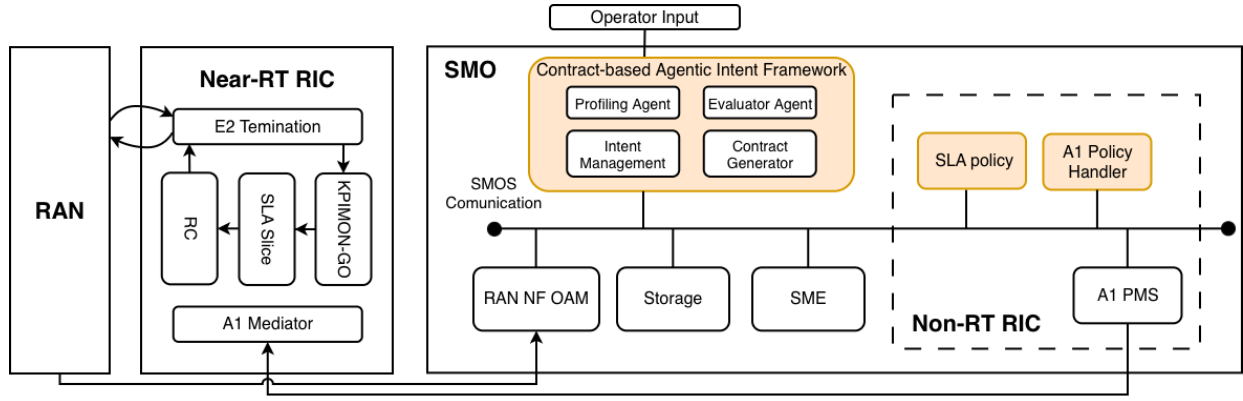


Fig. 2. System architecture of a CAIF

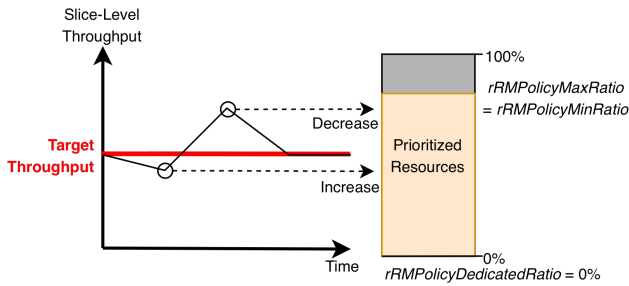


Fig. 3. Closed-Loop RAN Slice Control Algorithm

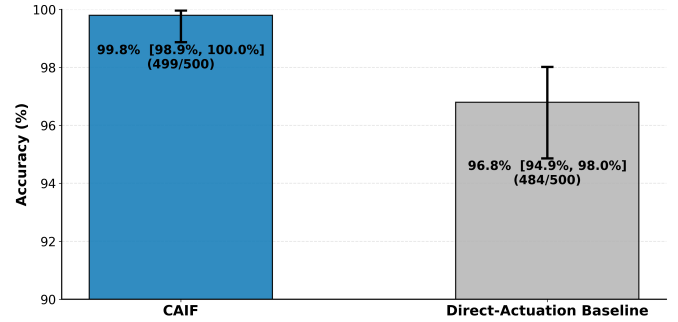


Fig. 4. Overall Accuracy with 95% Confidence Intervals

and parameter values, such as varying slice identifiers and throughput thresholds.

B. Semantic Integrity and Output Validation

This evaluation assesses the semantic integrity of the intent-to-policy translation process. We observed whether the output generated by the baseline LLM or CAIF can align with the original user intents, or if semantic deviation occurred during translation. As illustrated in Fig. 4, the baseline architecture (configured as Table II) achieves a translation accuracy of 96.8%. However, its 95% confidence interval ranges from 94.9% to 98.0%, indicating a higher risk of semantic inconsistency. From the 500 test cases, 16 errors were observed in the baseline results. Some of the errors identified as false positives, where the generated A1 policies were structurally valid yet semantically misaligned with the user's intent. This error category is particularly dangerous because these policies pass basic syntax checks and are therefore forwarded for execution, directly violating the operational requirements. The remaining failures were caused by formatting errors, where the output did not match the strict requirements of the rApp. In the baseline architecture, these malformed commands are sent directly to the network, leading to guaranteed execution errors.

In contrast, CAIF (configured as Table III) achieves an accuracy of 99.8%, with a much narrower 95% confidence

TABLE II
DIRECT-ACTUATION BASELINE LLM AGENT INFERENCE PARAMETERS

Parameter	Value
LLM Agent Model	llama-3.3-nemotron-super-49b-v1.5
Temperature	0.6
Top P	0.95

interval ranging from 98.9% to 100.0%, corresponding to 499 successful translations out of 500 intent instances. This reduced error margin indicates that CAIF provides more reliable validation and better preserves semantic consistency than the general LLM-based baseline.

TABLE III
CAIF LLM AGENT INFERENCE PARAMETERS

Parameter	Value
Profiling Agent Model	Qwen3-4B-Instruct-2507
Evaluator Agent Model	llama-3.3-nemotron-super-49b-v1.5
Temperature	0.6
Top P	0.95

We conduct a field-level accuracy analysis, as illustrated in Fig. 5. The results show that CAIF achieves 100% accuracy with 95% confidence interval ranging from 99.2% to 100% across several key intent attributes, which is much better than direct-actuation baseline. Fig. 6 presents a per-shot analysis

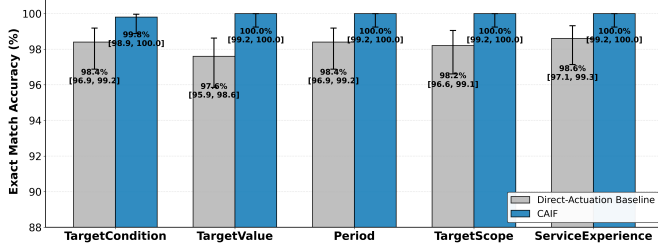


Fig. 5. Field-Level Accuracy with 95% Confidence Interval

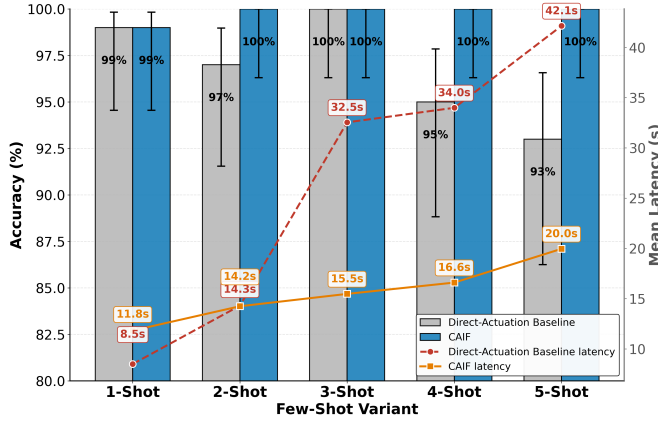


Fig. 6. Per-Shot Mean Latency & Accuracy with 95% Confidence Intervals

showing that the baseline performance degrades as more iteration required in order for system to fulfill KPIs from intent, particularly in the 4-shot and 5-shot scenarios. We tested the latency, due to the dual-agent coordination CAIF introduces a small overhead in the 1-shot scenario (11.8s vs. 8.5s), but the latency break-even occurs at the 2-shot setting, and CAIF consistently outperforms the baseline in the 3, 4 and 5-shot scenarios.

C. End-to-End Intent-Based Network in SLA Slice Use Case

To show the practical utility of the framework, we demonstrate the system in a realistic deployment scenario where the agent manages dynamic network slicing within a simulated environment. This test validates the ability of the contract-based logic to translate high-level intent into specific radio resource management actions within a complex campus network.

We simulate a campus deployment scenario using the RAN Simulator. This scenario requires three cellular networks consisting of two distinct slices: Enhanced Mobile Broadband (eMBB) and Massive Machine Type Communications (mMTC). The eMBB slice includes ten mobile users engaged in voice calls and thirty stationary users streaming video content. In contrast, the mMTC slice is configured with eighty IoT devices. Detailed network and user equipment parameters for this setup are provided in Table IV and Table V. To validate the End-to-End functionality from intent formulation to actual network response, we conducted two independent

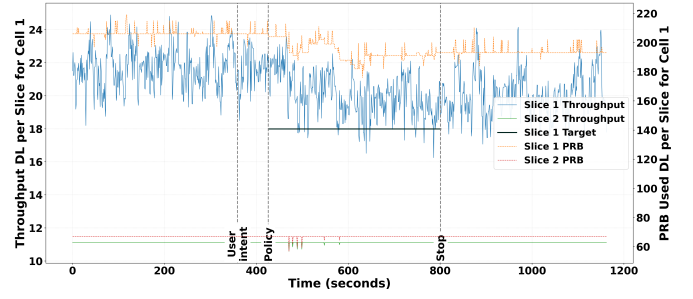


Fig. 7. Time series result for single intent assurance in slice 1 of cell 1

experiments. The first, Single Intent Assurance, verifies the translation of user intent into A1 policies and confirms that the Near-RT RIC closed loop mechanism is correctly triggered. The second, Dynamic Intent Assurance, evaluates the system stability when handling multiple intents arriving at different time intervals.

 TABLE IV
SCENARIO SIMULATION

Parameter	Value
Simulator	VIAVI AI RSG
Version	2.6
Number of gNB	3
Initial number of UE(s) at gNB ₁	43
Initial number of UE(s) at gNB ₂	31
Initial number of UE(s) at gNB ₃	46

1) *Single Intent Assurance Evaluation:* The single intent assurance results for slice 1 are illustrated in Fig. 7. When the operator submits a request to "decrease downlink throughput by 20% in 5 minutes" for slice 1, the system first determines that the current throughput is approximately 22 Mbps. It then calculates the 20% reduction to set a target of 18 Mbps and prepares the data for the A1 policy.

When Near-RT RIC receives the A1 Policy, xApps start adjusting the rRMPolicyMaxRatio and rRMPolicyMinRatio via E2 interface to maintain the throughput within the 18Mbps every second. After stopping the policy, rRMPolicyMaxRatio and rRMPolicyMinRatio values persist at the final control state and throughput starts to aggressively change.

2) *Dynamic Intent Assurance Evaluation:* The second test, shown in Fig. 8 illustrates how the system handles multiple requests from an operator to manage resources across slice 1 and slice 2. When the operator submits the first intent to "decrease downlink throughput by 20% in 10 minutes" for slice 1, the system identifies that the current throughput is approximately 22 Mbps. It then translates that 20% reduction into a target of 18 Mbps and prepares the data for the A1 policy.

When the Near-RT RIC receives the A1 policy (indicated by the vertical dashed line labeled "Policy 1"), xApps begin adjusting the rRMPolicyMaxRatio and rRMPolicyMinRatio via the E2 interface to maintain the throughput at 18 Mbps every second. Two minutes later, the operator submits another

TABLE V
UE AND SLICE SIMULATION

UE Group	Mobility Model	Quantity	Slice	Qos Identifier	Target Throughput	Description
Pedestrian	Random walk	10	eMBB (Slice 1)	1 (GBR)	0.5 Mbps	Conversation Voice
Person	Fixed	30	eMBB (Slice 1)	8 (Non-GBR)	40 Mbps	Video
IoT Device	Fixed	80	mMTC (Slice 2)	9 (Non-GBR)	0.25 Mbps	IoT Device

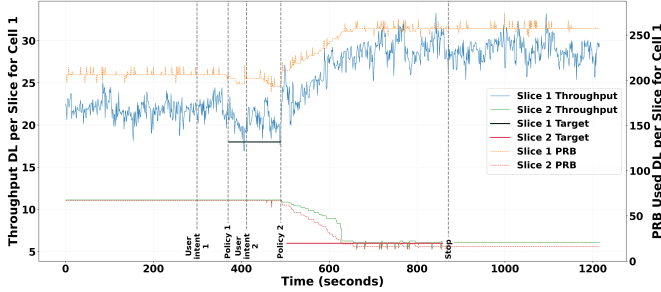


Fig. 8. Time series result for dynamic intent assurance from slice 1 to slice 2 of cell 1

request: "In slice ID 2 of cell 1, decrease downlink throughput by 50% in 5 minutes." The system identifies that the current throughput for slice 2 is approximately 12 Mbps, calculates the 50% reduction to a target of 6 Mbps, and prepares the data for the A1 policy.

When the Near-RT RIC receives this second A1 policy (indicated by the vertical dashed line labeled "Policy 2"), the SLA Slice xApp replaces the first policy. It then maintains slice 2 at 6 Mbps by dynamically adjusting the rRMPolicyMaxRatio and rRMPolicyMinRatio. After the policy stops (indicated by the vertical dashed line labeled "Stop"), the values for these ratios remain at their final controlled levels.

IV. CONCLUSIONS AND FUTURE WORKS

This study introduces CAIF, enforced across rApp and xApp workflows for slice resource allocation on an O-RAN testbed. From the experimental evaluation, we identified several critical insights that validate the architecture and motivate directions for future research. First, the results we obtained from our testing show contract-based approach combined with Generative AI for intent translation, significantly outperforms traditional baselines. It effectively acts as a guardrail to prevent minor semantic drifts or format errors can lead to policy execution failures. Future work will expand the evaluation dataset to explore more robust mechanisms for constraining LLM behavior. Second, the End-to-End SLA slice experiments demonstrate that the proposed framework operates effectively in simulated RAN environments with specific scenarios that reflect real-world situations. Both single and dynamic intent assurance scenarios confirm that the framework can continuously adapt network control policies while maintaining SLA requirements. Motivated by these results, future work will extend the proposed framework to support additional RAN management use cases, such as Energy Saving and Traffic Steering.

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REFERENCES

- [1] K. Abbas, T. A. Khan, M. Afaq, and W.-C. Song, "Network slice life-cycle management for 5g mobile networks: An intent-based networking approach," *IEEE Access*, vol. 9, pp. 80 128–80 146, 2021.
- [2] J. Zhang, Z. Liu, Y. Zhu, E. Shi, B. Xu, C. Yuen, D. Niyato, M. Debbah, S. Jin, B. Ai, Xuemin, and Shen, "Multi-agent reinforcement learning in wireless distributed networks for 6g," 2025. [Online]. Available: <https://arxiv.org/abs/2502.05812>
- [3] A. Mekrache, A. Ksentini, and C. Verikoukis, "Intent-based management of next-generation networks: an llm-centric approach," *IEEE Network*, vol. 38, no. 5, pp. 29–36, 2024.
- [4] M. Polese, L. Bonati, S. D'Oro, S. Basagni, and T. Melodia, "Coloran: Developing machine learning-based xapps for open ran closed-loop control on programmable experimental platforms," *IEEE Transactions on Mobile Computing*, vol. 22, no. 10, pp. 5787–5800, 2023.
- [5] M. A. Habib, H. Zhou, P. E. Iturria-Rivera, M. Elsayed, M. Bavand, R. Gaigalas, Y. Ozcan, and M. Erol-Kantarci, "Intent-driven intelligent control and orchestration in o-ran via hierarchical reinforcement learning," in *2023 IEEE 20th International Conference on Mobile Ad Hoc and Smart Systems (MASS)*, 2023, pp. 55–61.
- [6] Y. Xiao, G. Shi, and P. Zhang, "Toward agentic ai networking in 6g: A generative foundation model-as-agent approach," *IEEE Communications Magazine*, vol. 63, no. 9, pp. 68–74, 2025.
- [7] D. M. Manias, A. Chouman, and A. Shami, "Towards intent-based network management: Large language models for intent extraction in 5g core networks," in *2024 20th International Conference on the Design of Reliable Communication Networks (DRCN)*, 2024, pp. 1–6.
- [8] K. Dzevaroska, J. Lin, A. Tizghadam, and A. Leon-Garcia, "Llm-based policy generation for intent-based management of applications," in *2023 19th International Conference on Network and Service Management (CNSM)*, 2023, pp. 1–7.
- [9] F. A. Bimo, M. A. C. Galdon, C.-K. Lai, R.-G. Cheng, and E. K. P. Chong, "Intent-based network for ran management with large language models," 2025. [Online]. Available: <https://arxiv.org/abs/2507.14230>
- [10] TMFORUM, "Tmf921 intent management api user guide." [Online]. Available: <https://www.tmf921-intent-management-api-user-guide-v5-0-0/>
- [11] F. A. Bimo, R.-G. Cheng, C.-C. Tseng, C.-R. Chiang, C.-H. Huang, and X.-W. Lin, "Design and implementation of next-generation research platforms," in *2023 IEEE Globecom Workshops (GC Wkshps)*, 2023, pp. 1777–1782.
- [12] F. A. Bimo, C.-K. Lai, Z.-Y. Yang, and R.-G. Cheng, "Contract-based Agentic Intent Framework," GitHub, 2026, accessed: Feb. 28, 2026. [Online]. Available: https://github.com/bmw-ece-ntust/Contract-based_Agentic_Intent_Framework